

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'NG_FFCutsFits.mat');

if ~exist('NG_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    NG_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(NG_FFCutsFits);
nf = 11

```

```

nf =
11

```

```

%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off110NGminFF'

```

```

selectCuts =
'Ei800_off110NGminFF'

```

```

if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei800_off110NGminFF

```

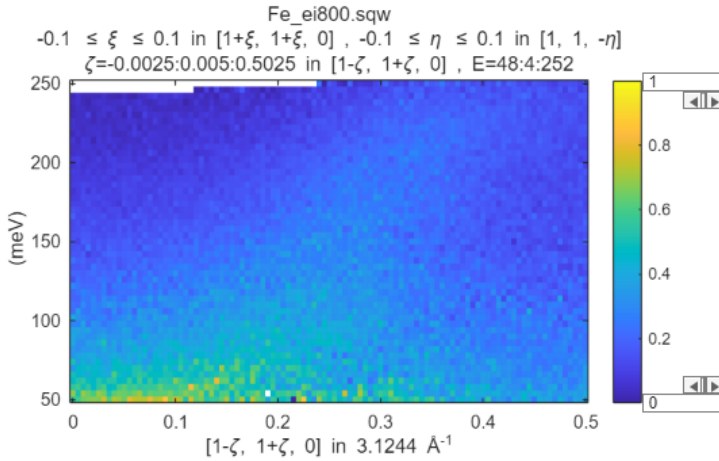
```

%selectCuts= 'Ei400_off200GPmaxFF';
cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample=IX_sample(true, [1,0,0], [0,1,0], 'cuboid', [0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "NG";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

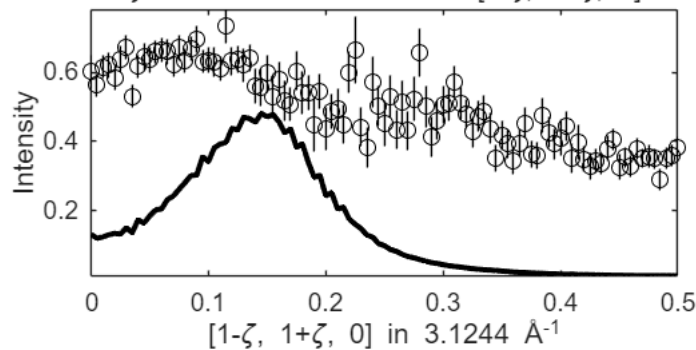
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

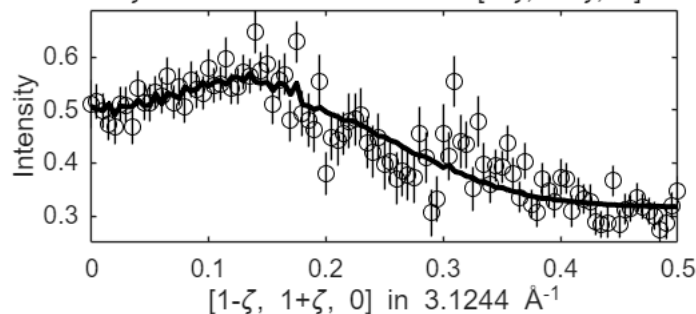
```
*****
**** En = 53±5 going to 243
*****
```

Fe_ei800.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $4\theta = 0.1$
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

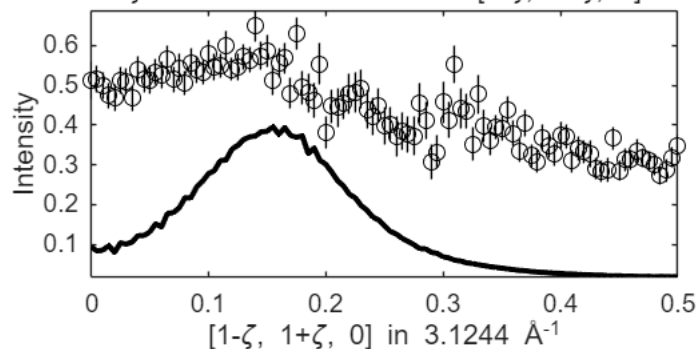


 **** En = 63±5 going to 243

Fe_ei800.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $5\theta = 0.1$
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



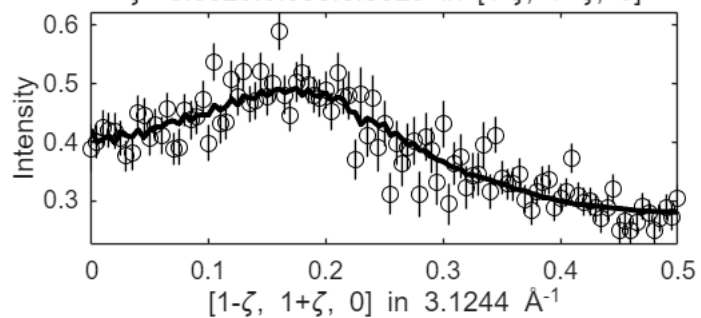
Fe_ei800.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, $5\theta = 0.1$
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



 **** En = 73±5 going to 243

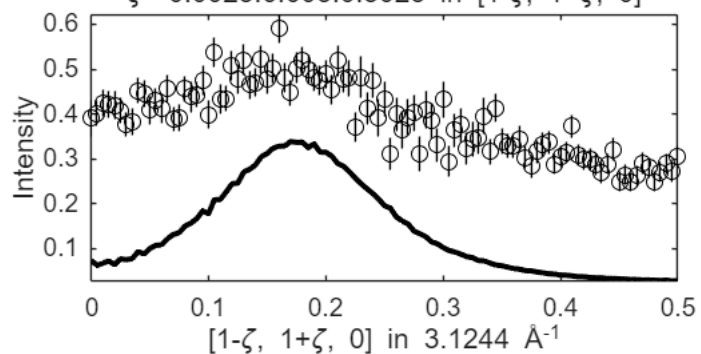
Fe_ei800.sqw

3.4 ± 0.16 ; $J_0 = 68 \pm 1.9$; $J_1 = 0 \pm 0$; $\gamma = 334.11 \pm 6$.
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 6ϵ
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



Fe_ei800.sqw

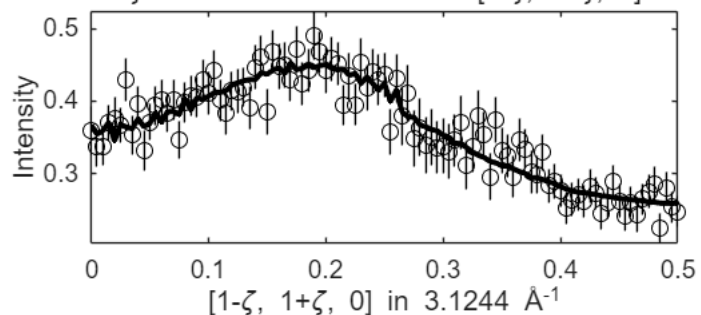
≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 6ϵ
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



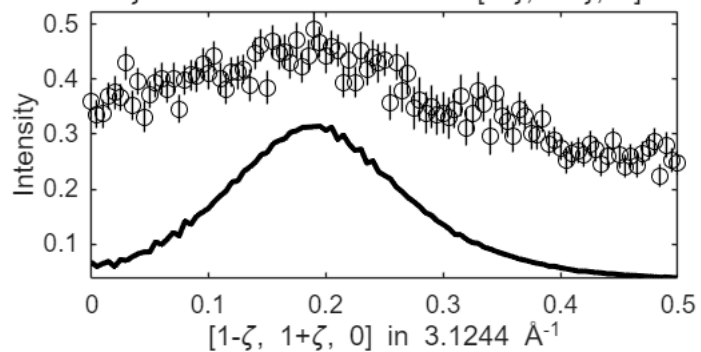
 **** En = 83 ± 5 going to 243

Fe_ei800.sqw

± 18 ; $J_0 = 83 \pm 2.8e+02$; $J_1 = 0 \pm 0$; $\gamma = 586.978$.
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 7ϵ
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

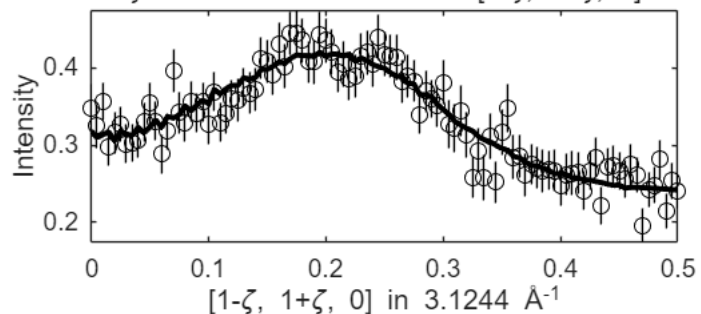


Fe_ei800.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 7%
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

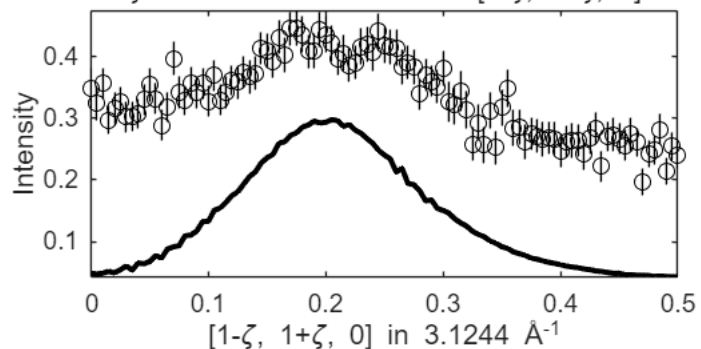


 **** En = 93±5 going to 243

Fe_ei800.sqw
 3 ± 7 ; $J_0 = 58\pm 97$; $J_1 = 0\pm 0$; $\gamma = 281.146\pm$
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 8%
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



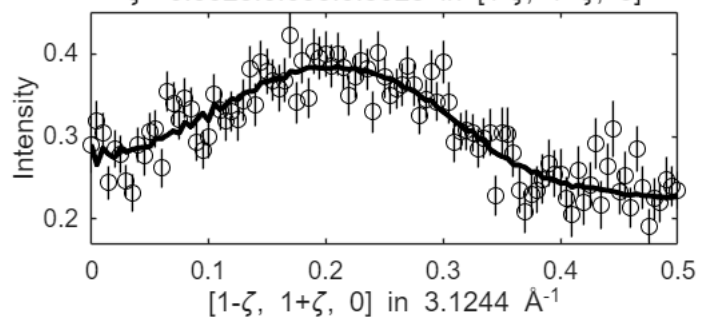
Fe_ei800.sqw
 ≤ 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 8%
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



 **** En = 103±5 going to 243

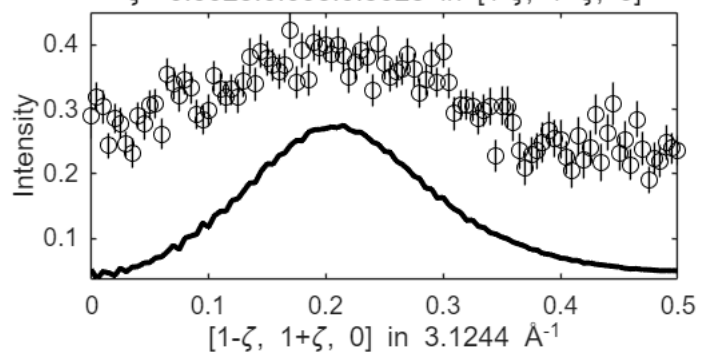
Fe_ei800.sqw

$J_0 = 78 \pm 1.3$; $J_1 = 0 \pm 0$; $\gamma = 543.659 \pm 9$
 $\xi = 0.1$ in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 98
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



Fe_ei800.sqw

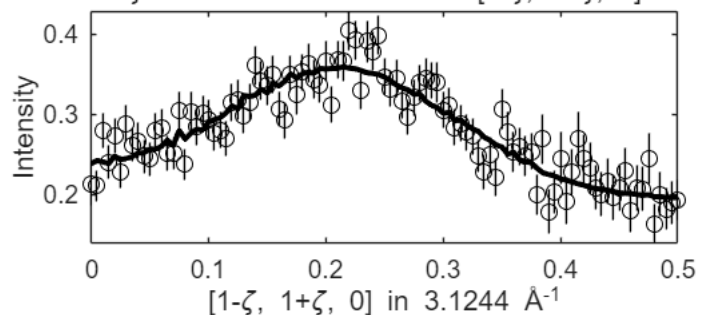
$\xi = 0.1$ in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 98
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



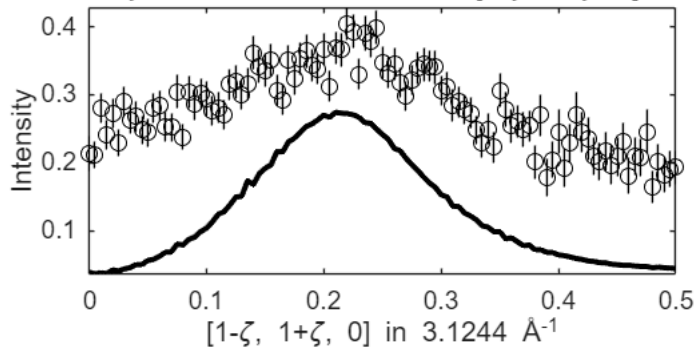
 **** En = 113±5 going to 243

Fe_ei800.sqw

$J_0 = 52 \pm 0.73$; $J_1 = 0 \pm 0$; $\gamma = 205.418 \pm 3$
 $\xi = 0.1$ in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 101
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

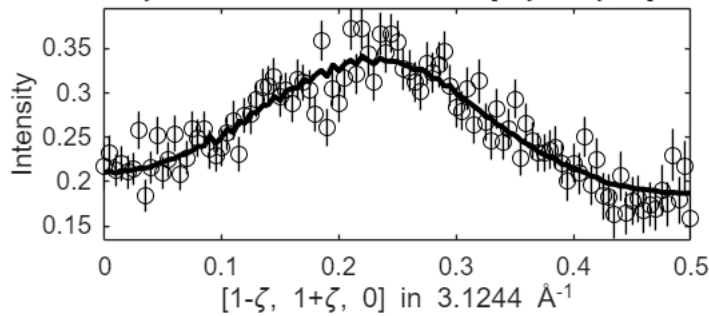


Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 10°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

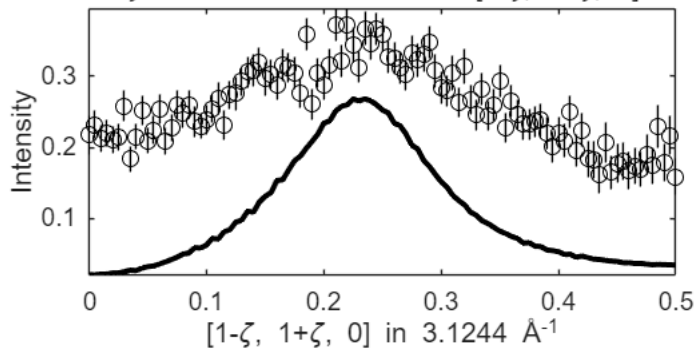


 **** En = 123±5 going to 243

Fe_ei800.sqw
 2 ± 0.047 ; $J_0 = 36 \pm 0.48$; $J_1 = 0 \pm 0$; $\gamma = 84.0729 \pm$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 11°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

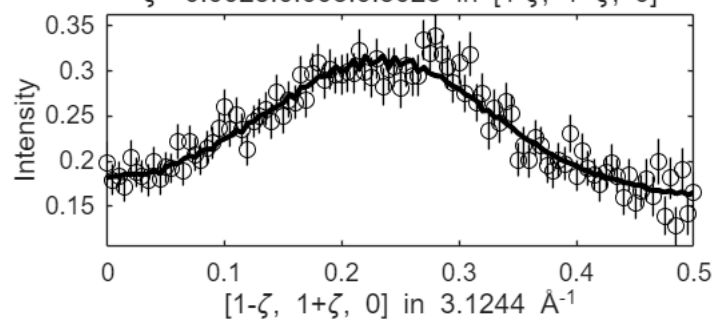


Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 11°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

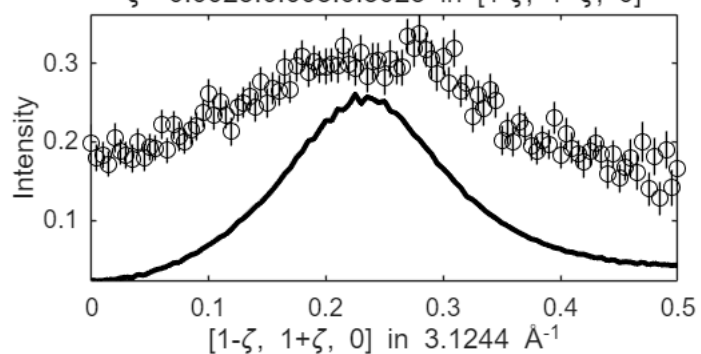


 **** En = 133±5 going to 243

Fe_ei800.sqw
 5 ± 0.052 ; $J_0 = 42 \pm 0.39$; $J_1 = 0 \pm 0$; $\gamma = 116.293 \pm 0.1$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 12°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

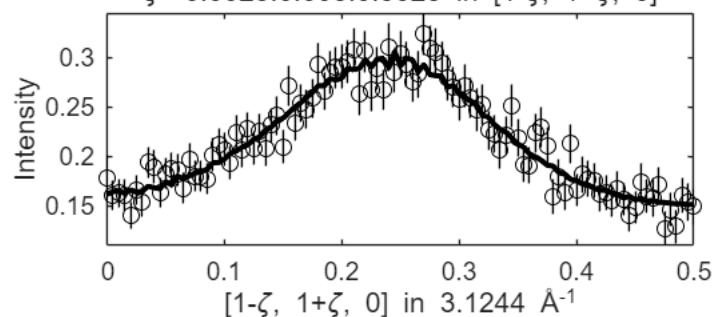


Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 12°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

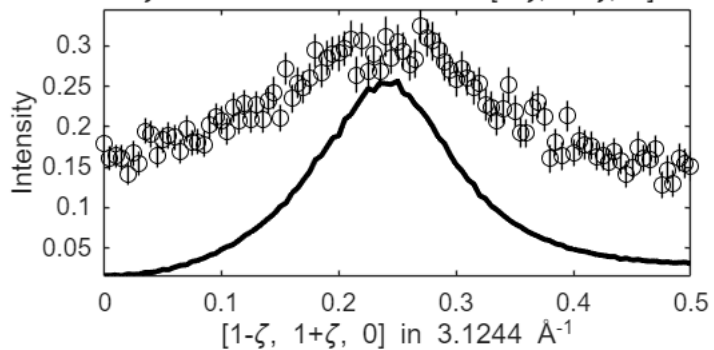


 **** $E_n = 143 \pm 5$ going to 243

Fe_ei800.sqw
 1 ± 0.033 ; $J_0 = 38 \pm 0.37$; $J_1 = 0 \pm 0$; $\gamma = 77.9788 \pm 0.1$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 13°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

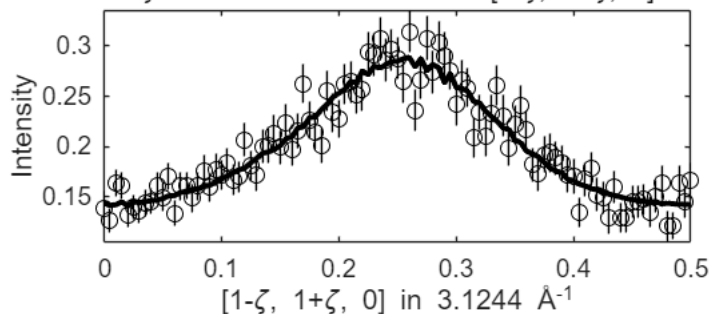


Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 134
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

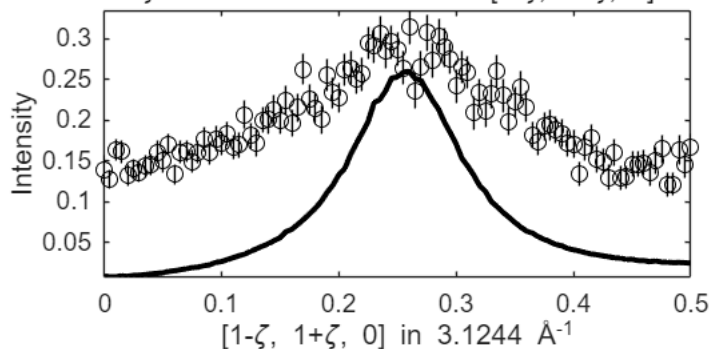


**** En = 153±5 going to 243

Fe_ei800.sqw
 78±0.11; J0= 36±0.93; J1= 0± 0; gamma=52.4399±1
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 144
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



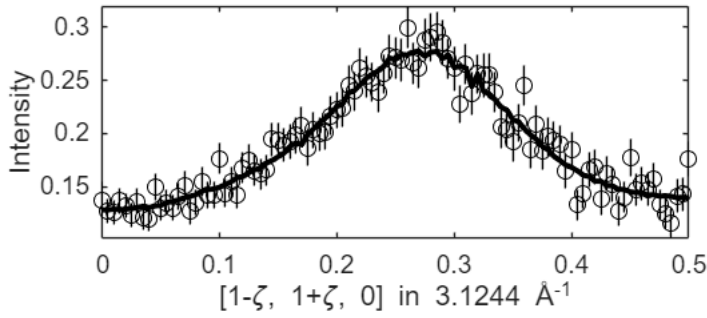
Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 144
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



**** En = 163±5 going to 243

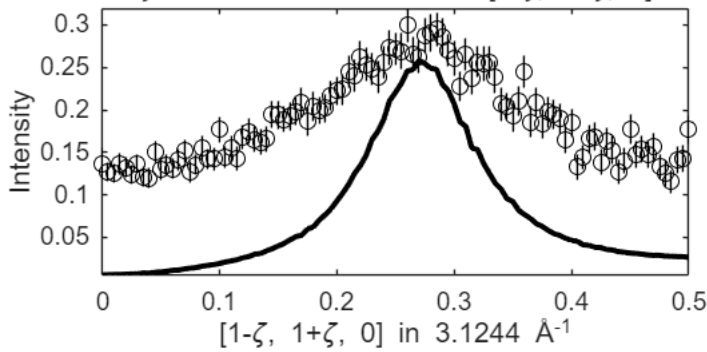
Fe_ei800.sqw

$\lambda \pm 0.017$; $J_0 = 34 \pm 0.27$; $J_1 = 0 \pm 0$; $\gamma = 46.6802 \pm 0.1$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 15°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



Fe_ei800.sqw

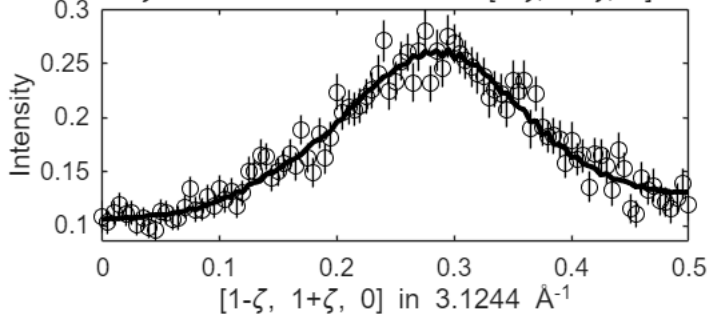
0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 15°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

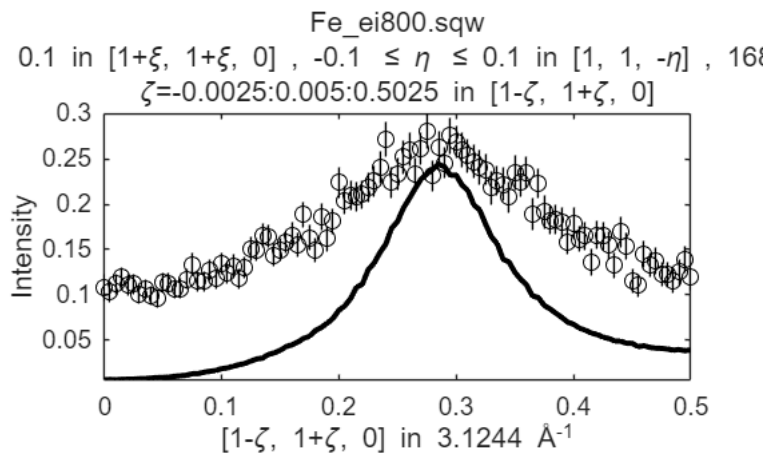


 **** $E_n = 173 \pm 5$ going to 243

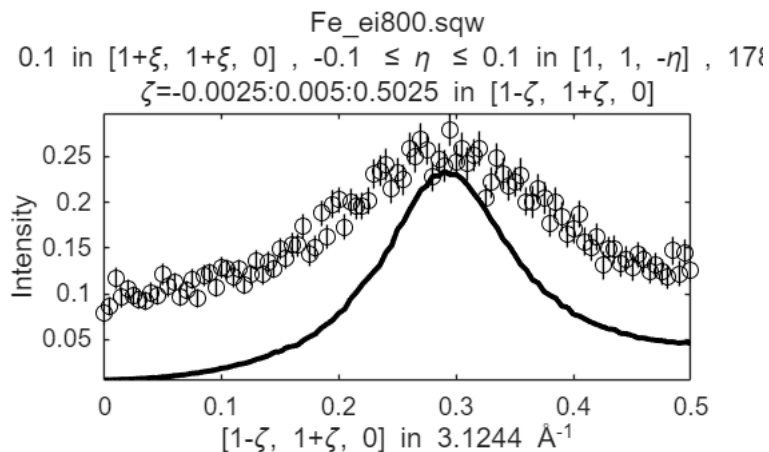
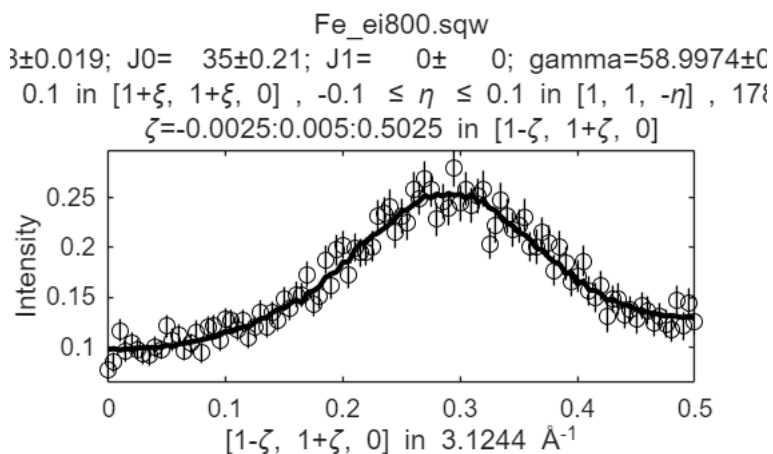
Fe_ei800.sqw

$\lambda \pm 0.018$; $J_0 = 35 \pm 0.23$; $J_1 = 0 \pm 0$; $\gamma = 54.3039 \pm 0.1$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 16°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$





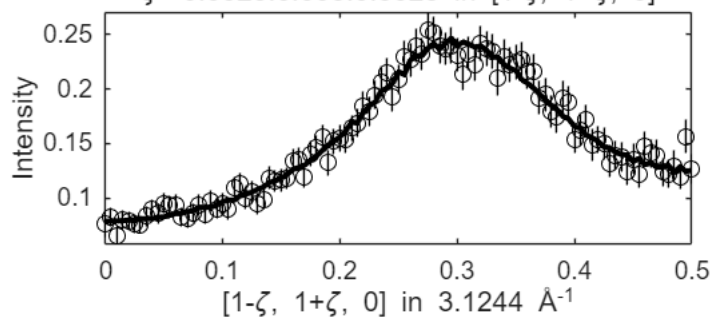
 **** En = 183 ± 5 going to 243



 **** En = 193 ± 5 going to 243

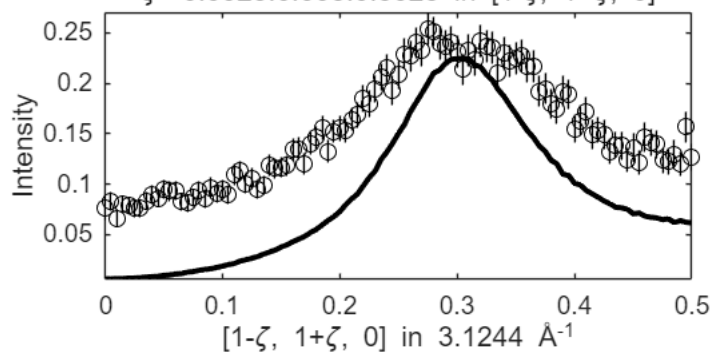
Fe_ei800.sqw

3 ± 0.019 ; $J_0 = 36 \pm 0.17$; $J_1 = 0 \pm 0$; $\gamma = 68.1188 \pm 0$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 18°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



Fe_ei800.sqw

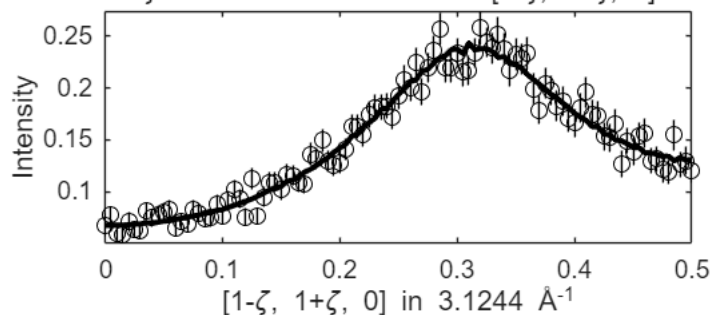
0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 18°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



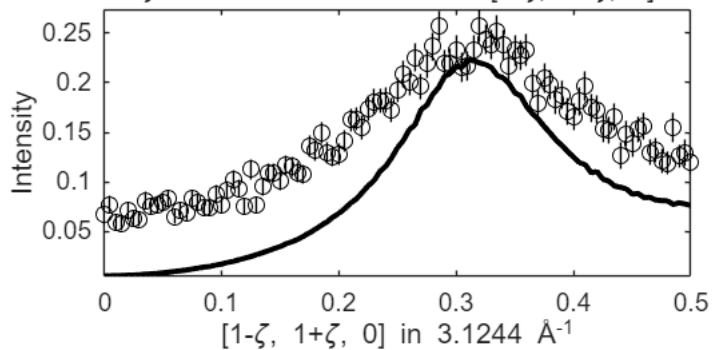
 **** $E_n = 203 \pm 5$ going to 243

Fe_ei800.sqw

3 ± 0.025 ; $J_0 = 36 \pm 0.21$; $J_1 = 0 \pm 0$; $\gamma = 73.0081 \pm$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 19°
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

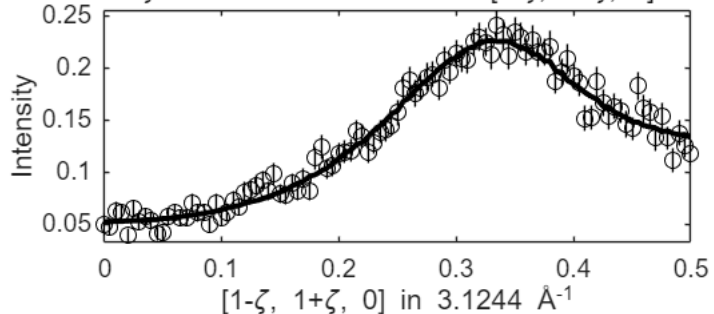


Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 194
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

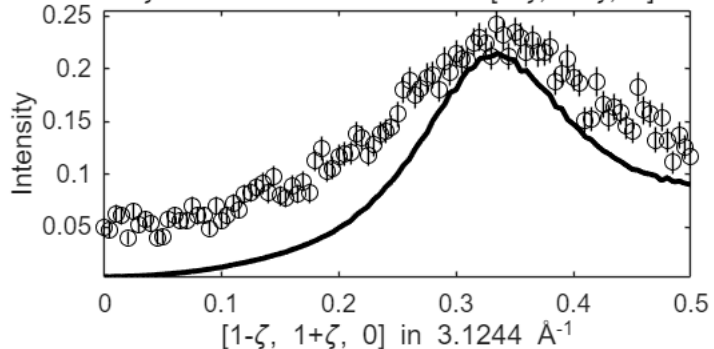


**** En = 213±5 going to 243

Fe_ei800.sqw
 9±0.024; J0= 36± 0.2; J1= 0± 0; gamma=71.9408±
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 204
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

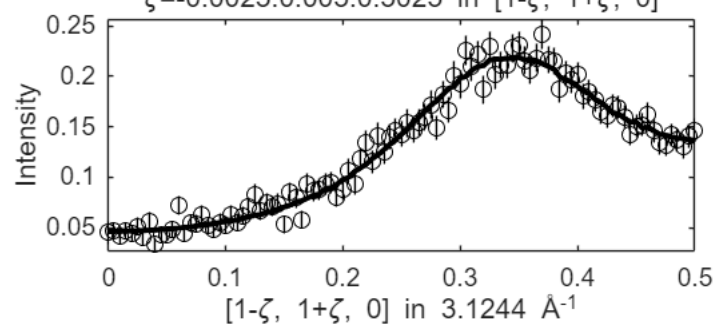


Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 204
 $\zeta=-0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

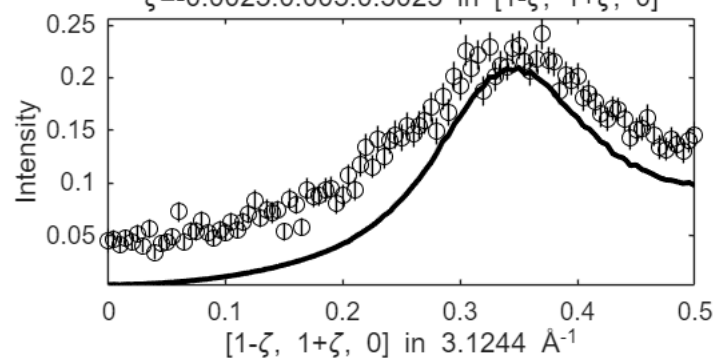


**** En = 223±5 going to 243

Fe_ei800.sqw
 9 ± 0.023 ; $J_0 = 36 \pm 0.18$; $J_1 = 0 \pm 0$; $\gamma = 70.6059 \pm 0.1$ in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 21θ
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

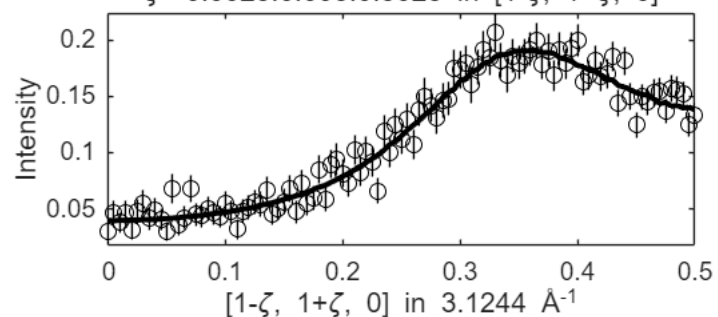


Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 21θ
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

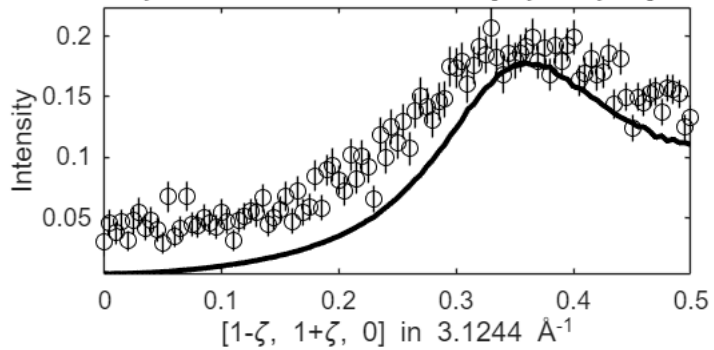


 **** $E_n = 233 \pm 5$ going to 243

Fe_ei800.sqw
 4 ± 0.033 ; $J_0 = 36 \pm 0.24$; $J_1 = 0 \pm 0$; $\gamma = 72.5911 \pm 0.1$ in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 22θ
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

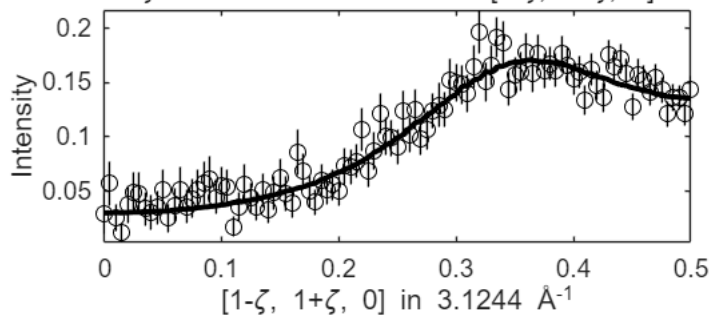


Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 224
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$

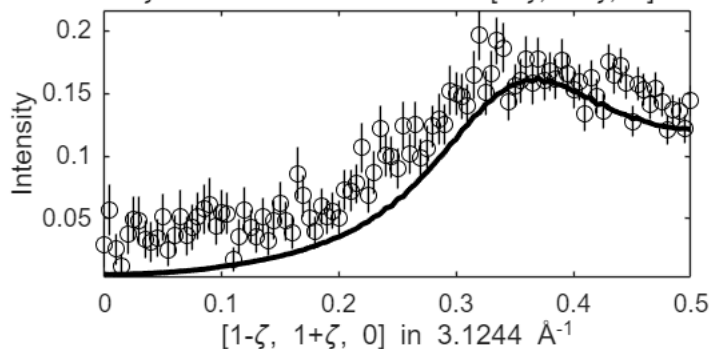


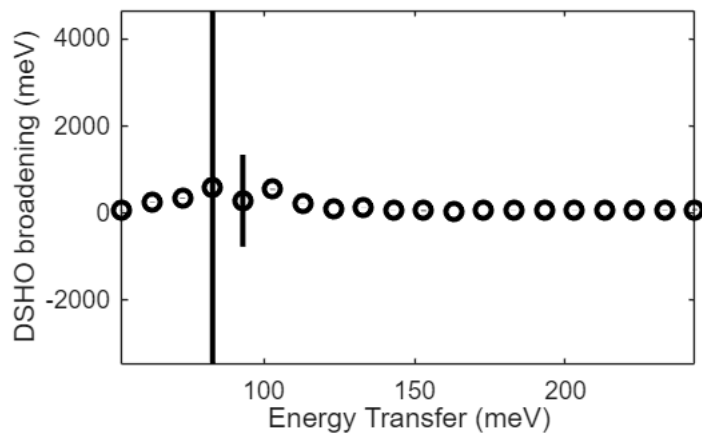
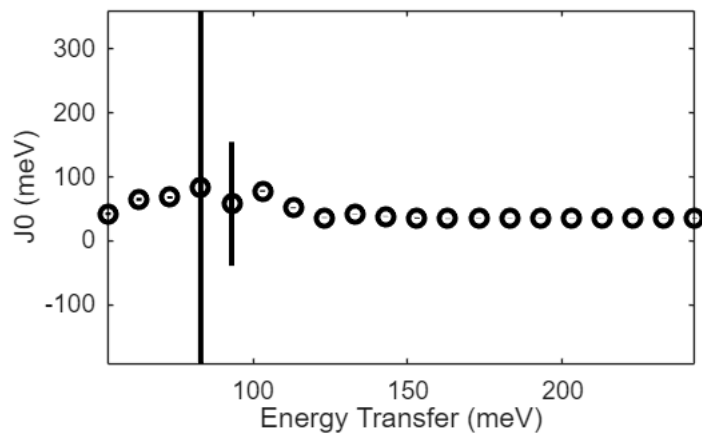
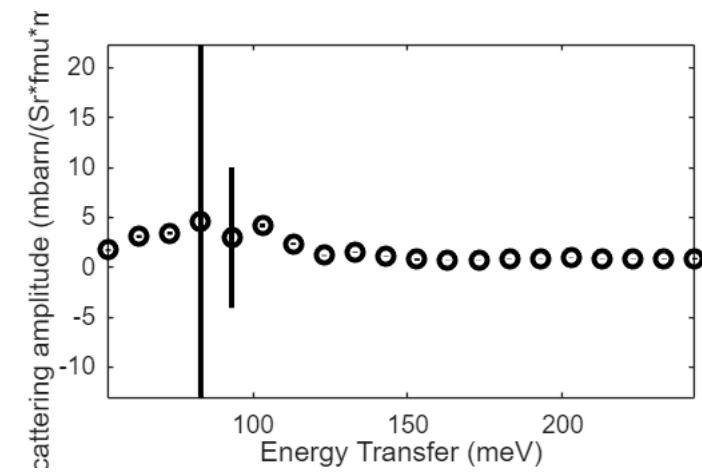
 **** En = 243±5 going to 243

Fe_ei800.sqw
 6 ± 0.083 ; $J_0 = 36 \pm 0.42$; $J_1 = 0 \pm 0$; $\gamma = 78.0507 \pm$
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 234
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$



Fe_ei800.sqw
 0.1 in $[1+\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, -\eta]$, 234
 $\zeta = -0.0025:0.005:0.5025$ in $[1-\zeta, 1+\zeta, 0]$





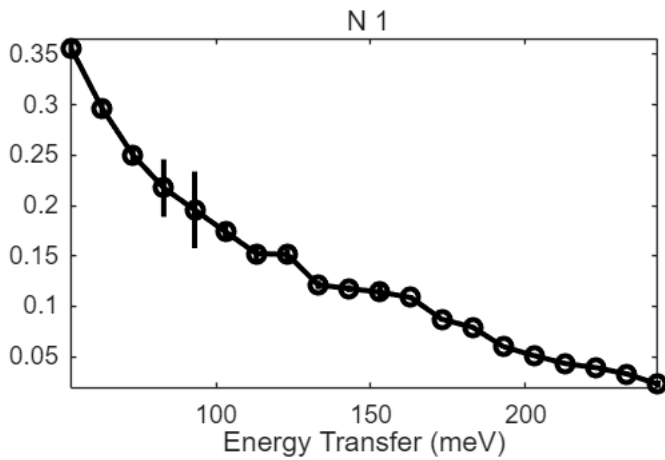
```

c1 = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [c1{:}];

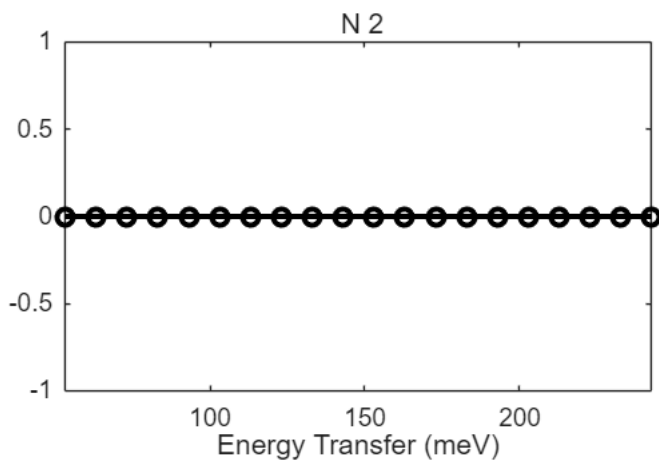
beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struct = NG_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```